

Lightweight ECG Signal Classification Using Machine Learning and Deep Learning Models

Ayush Nitin Deshmukh

MSc Data Science

Queen Mary University of London

Supervisor: Dr Tayyab Ahmad Ansari

Abstract—This paper investigates how electrocardiogram (ECG) recordings can be used for physiological-state classification while balancing predictive performance with computational efficiency. A software pipeline is evaluated on free-living wearable ECG using the Wake/Sleep supervision available in the selected Dryad dataset. Thirty participants and 58 ECG recordings were analysed. ECG was sampled at 250 Hz and divided into 10-second windows containing 2,500 samples with 50% overlap. Logistic Regression and Linear Support Vector Machine (SVM) were trained on nine statistical features, while a compact convolutional neural network (CNN) and an LSTM received the waveform directly. The CNN gave the strongest validation performance and was selected for subsequent pruning and post-training dynamic-range quantisation. Quantisation reduced measured model storage from 14.8789 KB to 9.8047 KB and reduced paired validation prediction time from 51.0936 s to 27.6914 s. On five held-out participants, the final quantised CNN achieved an F1-score of 0.7380 and ROC-AUC of 0.7736. These results indicate that a compact waveform model can distinguish the available Wake/Sleep labels while retaining useful predictive performance after quantisation. The small number of independent test participants and the temporal label-matching procedure limit broader generalisation claims.

Index Terms—ECG classification, physiological-state classification, Wake/Sleep classification, lightweight CNN, model compression, pruning, quantisation

I. INTRODUCTION

Electrocardiogram (ECG) signals provide a continuous source of physiological information that can be analysed using machine learning (ML) and deep learning (DL). The growing use of wearable ECG recording makes it important to consider not only predictive performance, but also the computational requirements of the models used to analyse these signals. Automated ECG analysis spans conventional signal processing, feature-based ML and end-to-end DL approaches (Stracina et al., 2022; Petmezas et al., 2022; Chen et al., 2022; Safdar et al., 2024). ECG recordings can also contain baseline wander, power-line interference, movement artefacts and muscle noise, making signal preprocessing and evaluation design important parts of the modelling pipeline (Jia et al., 2024; Safdar et al., 2024).

Physiological-state classification provides an important application of ECG analysis beyond direct waveform interpretation. Choi and Sung demonstrated single-channel ECG-based sleep-stage classification using an end-to-end neural architecture (Choi and Sung, 2023). Sun et al. similarly showed that ECG and respiratory signals can support sleep-state classification using convolutional and recurrent neural networks. Importantly

for the present work, their discussion noted that using the full ECG waveform could potentially improve performance compared with using R-peak information, while also raising concerns about robustness across different recording devices (Sun et al., 2020). This limitation creates a useful methodological direction: the benefit of retaining waveform information needs to be considered alongside the computational cost of the resulting model.

A second strand of research has focused on making ECG models more efficient. Lightweight CNNs, quantised neural networks and pruning methods have demonstrated that model storage and computational requirements can be reduced (Kim et al., 2022; De Melo Ribeiro et al., 2022; Liu et al., 2024). However, these studies evaluate efficiency in different datasets, tasks and execution environments. The systematic ECG literature also shows substantial variation in data sources, representations, participant splitting and evaluation procedures, making direct comparison between reported results difficult (Petmezas et al., 2022; Xiao et al., 2023; Avula et al., 2023).

Research gap. Sun et al. (2020) leave an important representation-level question open. Their study used R-peak information and noted that the full ECG waveform might provide additional information and improve performance, while potentially reducing robustness when signals are obtained from different or simpler devices (Sun et al., 2020). This suggests that the choice between a compact ECG representation and the full waveform remains an important practical consideration for physiological-state classification. At the same time, work on lightweight ECG models has shown that established techniques such as pruning and quantisation can reduce model size and computational requirements (Kim et al., 2022; De Melo Ribeiro et al., 2022; Liu et al., 2024). These two lines of work, however, have not been brought together in the same participant-level experiment for the free-living wearable ECG setting considered here.

The gap addressed by this paper is therefore not simply the use of a different dataset. It is the lack of an integrated evaluation of representation choice and model efficiency: whether a waveform-based model provides a useful advantage over compact statistical ECG features for the available physiological-state task, and whether the selected waveform model can subsequently be compressed without a substantial loss of predictive performance. This paper addresses the gap by comparing feature-based ML with waveform-based DL under

the same participant-level partitions, selecting the strongest baseline using validation data, and then evaluating pruning and post-training quantisation within the same pipeline.

The available supervised annotation in the selected Dryad recordings is Wake/Sleep, with 1 representing Awake and 0 representing Asleep (Stokes et al., 2025). The research problem is therefore to evaluate how feature-based ML and waveform-based DL behave for this task and whether the strongest waveform model can be made smaller and faster through validation-driven compression.

The contributions of this paper are:

- A participant-level comparison of Logistic Regression and Linear SVM using compact statistical ECG features against a compact CNN and LSTM using the ECG waveform directly.
- A validation-driven model-selection procedure in which the strongest baseline is selected before compression and the held-out participants remain untouched until the final evaluation.
- An experimental evaluation of magnitude-based pruning and post-training dynamic-range quantisation within the same ECG classification pipeline, measuring both predictive performance and software-level efficiency.
- A participant-level assessment of the final quantised CNN on previously unseen participants, with the efficiency results interpreted as software measurements rather than evidence of deployment on a particular wearable processor.

II. BACKGROUND AND RELATED WORK

A. ECG preprocessing and evaluation

ECG preprocessing is commonly used to reduce unwanted signal components before downstream analysis. Safdar et al. reviewed preprocessing and AI-based ECG analysis across classical signal processing, ML and DL approaches, while Jia et al. reviewed ECG and magnetocardiography noise sources and denoising methods (Safdar et al., 2024; Jia et al., 2024). These reviews provide the general signal-processing rationale for the preprocessing stage in this paper; they do not prescribe the exact filter parameters used here.

The present implementation therefore treats the filtering parameters as methodological choices made for the Dryad recordings. A fourth-order Butterworth band-pass filter from 0.5 to 40 Hz was applied with zero-phase second-order-section filtering. The lower and upper cut-offs were selected to reduce slow baseline variation and higher-frequency noise while retaining the main ECG waveform components. The exact implementation, windowing procedure and quality-valid segment handling are reported in Section III rather than being presented as a reproduction of either review.

Evaluation design is also important when ECG results are compared across studies. Systematic reviews report differences in patient or participant splitting, preprocessing, data sources and performance reporting (Xiao et al., 2023; Avula et al., 2023; Petmezas et al., 2022). The present paper therefore uses participant-level train, validation and test groups so that

overlapping windows from the same participant do not cross partition boundaries.

B. Physiological-state classification from ECG

ECG can contain information related to physiological state as well as cardiac electrical activity. Choi and Sung demonstrated single-channel ECG-based sleep-stage classification using an end-to-end neural architecture (Choi and Sung, 2023). Sun et al. used ECG and respiratory signals with CNN and LSTM models for sleep-state classification and discussed the potential benefits and trade-offs of using richer ECG representations (Sun et al., 2020). The present experiment is narrower: it uses the Wake/Sleep supervision available in the Dryad recordings and compares feature-based ML with waveform-based DL before applying model compression.

C. Lightweight models, pruning and quantisation

The efficiency part of the work is informed by several related ECG studies. Kim et al. developed a lightweight CNN for low-power wearable ECG analysis, while De Melo Ribeiro et al. investigated quantised deep neural networks for real-time ECG monitoring (Kim et al., 2022; De Melo Ribeiro et al., 2022). Liu et al. investigated pruning and fine-tuning of a lightweight ECG classification network, showing that model reduction can be evaluated alongside predictive performance (Liu et al., 2024). These studies demonstrate the value of compact models, but their datasets, target tasks and execution environments differ from the free-living Wake/Sleep setting used here.

The present experiment brings model comparison and compression into one participant-level pipeline. Feature-based ML and waveform-based DL are first compared using the same participant-level partitions; the strongest baseline is then selected using validation data before pruning and quantisation are evaluated. The purpose is therefore to measure the behaviour of established techniques in the selected experimental setting rather than to introduce a new compression algorithm.

III. DATASET AND METHODOLOGY

A. Dataset selection and task definition

The Dryad “24-hour physiological monitoring” dataset was selected because it provides free-living wearable ECG recordings from 30 volunteers together with physiological measurements and Wake/Sleep annotations (Stokes et al., 2025). This makes it suitable for a software-only investigation of ECG-based physiological-state classification and model efficiency. The original project direction considered arrhythmia classification, but dataset inspection established that the available supervised target is Wake/Sleep. This defines the classification task used throughout this paper.

Initial exploration identified 30 participant folders and 58 ECG recordings. The ECG component was sampled at 250 Hz. A representative recording contained more than 21 million samples, motivating window-based processing rather than whole-recording modelling. The ECG files expose timestamp and waveform fields.

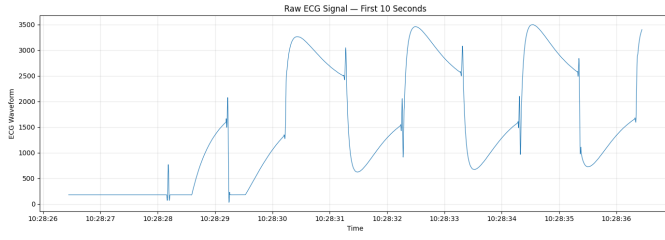


Fig. 1. Representative raw ECG waveform from participant 030 during dataset exploration. The figure shows the continuous waveform used as the starting point for subsequent preprocessing.

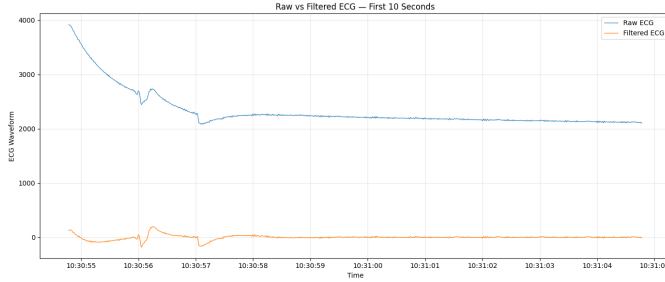


Fig. 2. Raw and filtered ECG over a representative 10-second interval. Filtering suppresses slow baseline variation and higher-frequency components while retaining the main ECG waveform.

The supplied `Output_ECG_Segmentor_data` was not used as predictive input. It is a separate derived representation associated with the ECG segmentation tool described by Haleem and Pecchia (Haleem and Pecchia, 2022). The modelling pipeline instead starts from the waveform itself so that the two model families can be compared from the same underlying ECG source.

The available supervised annotation was Wake/Sleep, with 1 representing Awake and 0 representing Asleep (Stokes et al., 2025). This annotation defines the target used for model training and evaluation.

B. ECG quality assessment and preprocessing

The sampling interval was estimated from consecutive timestamps. The median interval was 0.004 s, corresponding to 250 Hz. A frequency-domain check found negligible power around 50 Hz, so an additional notch filter was not included. This decision was made from the inspected signal rather than assumed from a generic preprocessing recipe.

A fourth-order Butterworth band-pass filter with 0.5–40 Hz cut-offs was implemented using zero-phase second-order-section filtering. The filtering stage was applied separately to continuous quality-valid segments so that invalid gaps were not bridged by the filter. The general purpose of the filtering stage is consistent with ECG preprocessing literature (Safdar et al., 2024; Jia et al., 2024); the specific parameters are part of the present implementation.

Each ECG window contained 10 s of data, equivalent to 2,500 samples at 250 Hz. Consecutive windows overlapped by

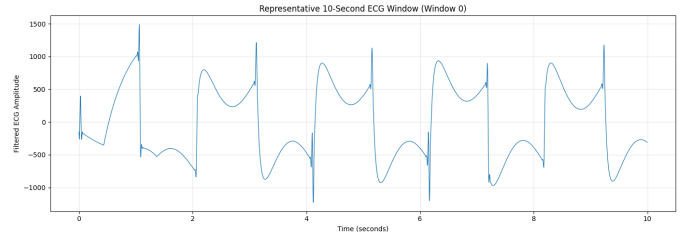


Fig. 3. Representative 10-second ECG window after filtering and segmentation. Each retained window contains 2,500 samples at the 250 Hz sampling rate.

50%, giving a 5 s step. Only complete windows located inside continuous quality-valid segments were retained.

C. Wake/Sleep temporal alignment

The Wake/Sleep spreadsheet contained 1,623 observations. Its date fields required explicit day/month/year parsing before alignment; after correction, all 1,623 timestamps were successfully reconstructed. ECG and annotation participant identifiers were then matched before temporal alignment.

For each candidate ECG window, the centre timestamp was matched to the nearest Wake/Sleep observation from the same participant. A label was retained only when the nearest observation was within a ± 15 -minute tolerance. This operational rule provided enough labelled windows for model development while keeping the matching procedure explicit. The final modelling dataset contained 421,166 windows.

D. Feature engineering and the ML–DL comparison

Two input pathways were deliberately maintained because the purpose of the ML–DL comparison was to examine different representations of the same ECG signal. The traditional ML pathway reduces each filtered window to a compact nine-feature representation. The DL pathway retains the complete waveform, allowing the neural models to learn a representation from the 2,500 samples.

For traditional ML, nine time-domain statistics were calculated from each filtered ECG window: mean, standard deviation, minimum, maximum, range, median, interquartile range, root mean square and signal energy. The resulting features were standardised with a `StandardScaler` fitted only on the training participants.

For DL, the complete 2,500-sample waveform was retained. Each window was standardised independently using its own mean and standard deviation. This preprocessing difference means that the comparison is between complete modelling pipelines, not a controlled experiment in which all four models receive an identical input representation.

E. Participant-level partitioning

The 30 participants were divided into non-overlapping training, validation and test groups. The final split contained 21 training participants, 4 validation participants and 5 held-out test participants. The test set contained 65,581 windows. Participant-level splitting was used so that overlapping windows from the same individual could not appear in different partitions.

TABLE I
BASELINE MODEL PERFORMANCE FROM THE FROZEN VALIDATION AND
HELD-OUT TEST EVALUATIONS.

Model	Val. F1	Val. AUC	Test F1	Test AUC
Logistic Regression	0.1403	0.5341	0.3589	0.6220
Linear SVM	0.1305	0.5337	0.3421	0.6248
Lightweight CNN	0.7932	0.7464	0.7423	0.7741
LSTM	0.4947	0.5695	0.4479	0.5498

F. Baseline models

Four models were evaluated. Logistic Regression and Linear SVM operated on the nine engineered features. The CNN and LSTM operated on the 2,500-sample waveform. The CNN was deliberately compact and contained 3,809 trainable parameters.

For the Logistic Regression baseline, the probability of the positive class is modelled as

$$P(y = 1 | \mathbf{x}) = \frac{1}{1 + \exp[-(\beta_0 + \beta^T \mathbf{x})]}, \quad (1)$$

where $\mathbf{x}$ is the vector of nine standardised ECG features, β contains the learned feature coefficients, and β_0 is the intercept. The output probability was converted to the predicted class using the selected decision threshold.

The LSTM was retained as a comparison model rather than assuming that recurrent modelling would necessarily improve performance. The CNN used convolutional layers followed by global average pooling, dropout and a sigmoid output. Training used Adam with a learning rate of 0.001 and binary cross-entropy. Class weighting was used for the neural models to reduce the influence of the observed class imbalance. No separate hyperparameter search is reported for the Logistic Regression or Linear SVM baselines; their validation scores therefore reflect the fixed configurations used in this experiment. This is an important qualification when interpreting their low F1-scores, because representation loss and model configuration cannot be separated completely from the present results.

G. Decision for model optimisation

The CNN was selected for optimisation because it produced the strongest validation F1-score and ROC-AUC among the four baseline models. This decision was made using validation data only; the held-out participants were not used to choose the model or compression level. The optimisation stage therefore follows the experimental sequence: compare representations and baseline models, select the strongest candidate, evaluate compression on validation data, and then perform one final held-out test.

H. Model optimisation

The selected CNN was frozen before optimisation. Magnitude-based unstructured pruning was evaluated at target sparsities of 20%, 40%, 60% and 80%. These experiments were performed using validation data only. Post-training dynamic-range quantisation was then applied to the frozen Float32 CNN using TensorFlow Lite. The quantised model

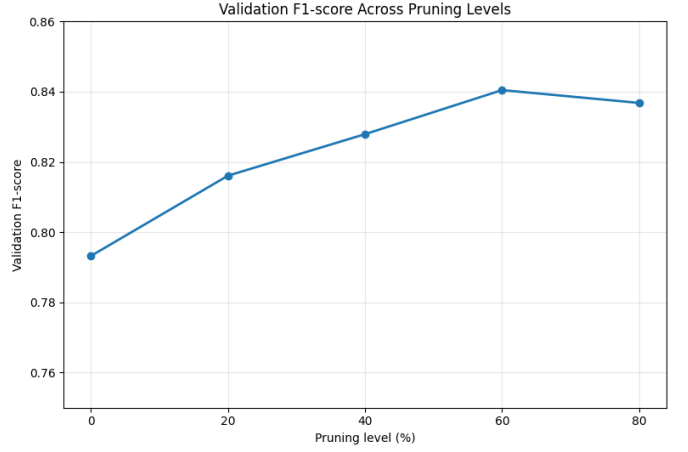


Fig. 4. Validation F1-score for the CNN at the evaluated pruning targets. The 60% target produced the highest validation F1 among the tested levels.

was evaluated using the same validation prediction and paired timing procedure used for the efficiency comparison.

The purpose of this stage is not to claim a new pruning or quantisation algorithm. Instead, it evaluates how established compression techniques behave within the particular Dryad Wake/Sleep pipeline. This distinguishes the present methodology from prior ECG compression studies such as Liu et al. and Ribeiro et al., which used different datasets, tasks and experimental environments (Liu et al., 2024; De Melo Ribeiro et al., 2022).

IV. EXPERIMENTAL RESULTS

A. Baseline model comparison

Table I summarises the baseline model comparison. The lightweight CNN achieved the strongest validation F1-score and ROC-AUC among the four models. Logistic Regression and Linear SVM produced validation ROC-AUC values close to chance level, while the LSTM remained below the CNN.

The CNN was therefore selected as the optimisation candidate using validation evidence only. The comparison should be interpreted as a comparison of the implemented feature-based and waveform-based pipelines, rather than as proof that CNN architecture is universally superior to traditional ML.

B. Pruning and quantisation

Pruning did not improve F1 monotonically as target sparsity increased. The 60% target produced the highest validation F1, with an actual measured sparsity of 58.73%, whereas the 80% target produced the highest validation ROC-AUC among the pruning levels. Increasing nominal sparsity therefore did not automatically produce the best predictive behaviour.

The frozen Float32 CNN occupied 14.8789 KB of weight storage. Dynamic-range quantisation reduced the model to 9.8047 KB, corresponding to a measured 34.10% reduction. Paired validation prediction time decreased from 51.0936 s to 27.6914 s, a measured reduction of 45.80% and a 1.845 $\times$ speed-up.

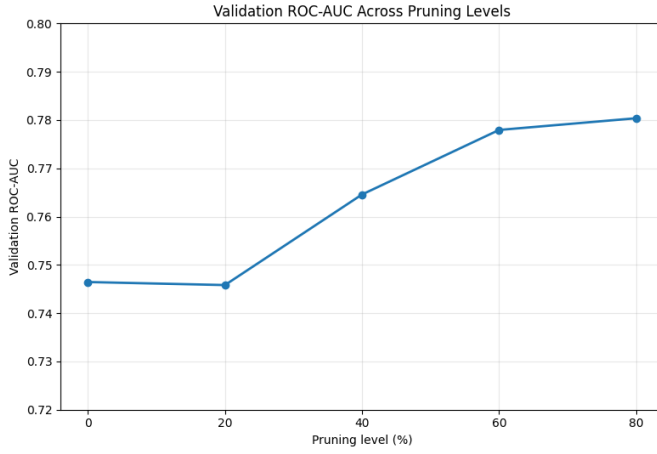


Fig. 5. Validation ROC-AUC for the CNN at the evaluated pruning targets. The highest ROC-AUC was observed at the 80% target.

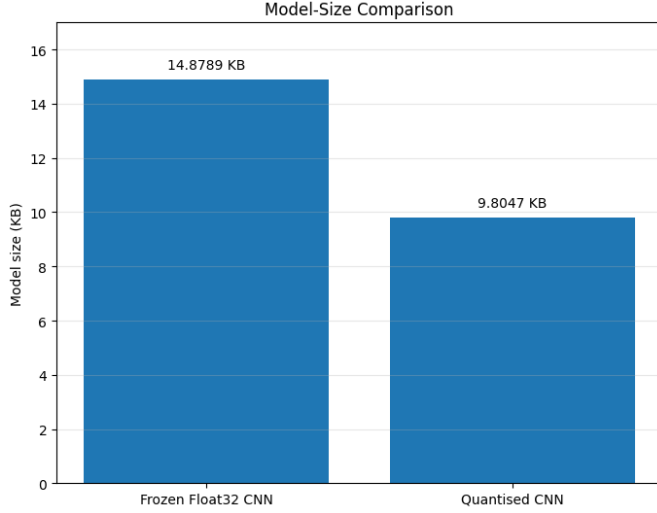


Fig. 6. Measured weight-storage size of the frozen Float32 CNN and the post-training dynamic-range quantised CNN. Quantisation reduced storage from 14.8789 KB to 9.8047 KB.

The quantised model retained validation $F1 = 0.7888$ and $ROC-AUC = 0.7447$, compared with 0.7932 and 0.7464 for the Float32 baseline. The small validation changes were therefore accompanied by measurable reductions in storage and prediction time.

C. Final held-out evaluation

After model selection was completed, the quantised CNN was evaluated once on the held-out test participants. The final test set contained 65,581 windows from five participants. At a frozen decision threshold of 0.5, the model achieved Accuracy = 0.6569, Precision = 0.8993, Recall = 0.6258, $F1 = 0.7380$ and $ROC-AUC = 0.7736$. The final prediction time was 46.1995 s.

The final confusion matrix contained 11,375 true negatives, 3,548 false positives, 18,956 false negatives and 31,702 true positives. The class-specific results indicate stronger perfor-

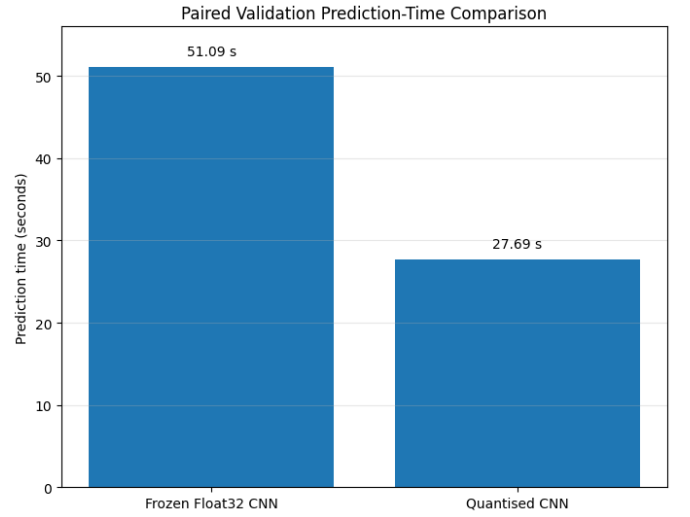


Fig. 7. Paired validation prediction time for the frozen Float32 CNN and the quantised CNN. The measured time decreased from 51.0936 s to 27.6914 s.

TABLE II
FINAL HELD-OUT TEST PERFORMANCE OF THE SELECTED QUANTISED CNN.

Metric	Quantised CNN
Accuracy	0.656852
Precision	0.899348
Recall	0.625804
F1-score	0.738045
ROC-AUC	0.773637
Test windows	65,581
Test participants	5
Prediction time (s)	46.199511

mance for class 1 than class 0, which should be considered alongside the imbalance in the labelled data.

V. DISCUSSION

The experiments show a clear difference between the feature-based and waveform-based pipelines. The validation $F1$ -score of the CNN was 0.7932, compared with 0.1403 for Logistic Regression and 0.1305 for Linear SVM. A reasonable interpretation is that the waveform representation retained temporal and morphological information that was lost when each window was reduced to nine summary statistics. This interpretation is consistent with broader ECG DL literature describing the use of learned waveform representations (Petmezas et al., 2022; Ansari et al., 2023; Prakash and Atef, 2025). The result should nevertheless be read as evidence for the implemented pipelines rather than as evidence that CNNs are universally superior to traditional ML.

The LSTM result was weaker than the CNN in this experiment. This does not establish that LSTMs are unsuitable for ECG. The experiment does not isolate whether the difference arose from the recurrent architecture, its training configuration, the input representation or the 10-second window design. The

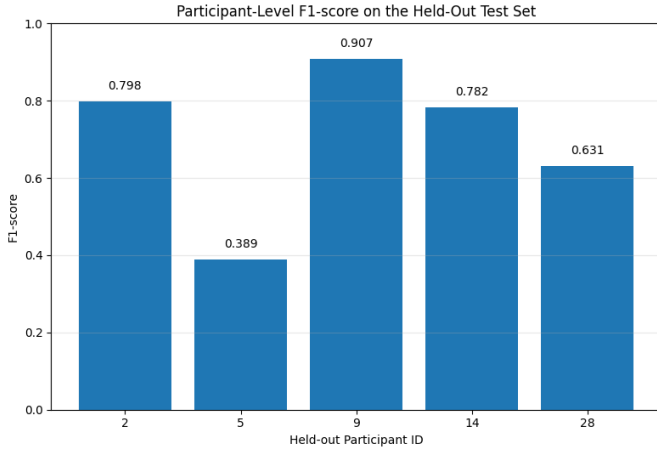


Fig. 8. F1-score for each held-out participant. The figure illustrates variation in performance across the five independent test participants.

result therefore supports selecting the CNN for this experiment, but not a universal ranking of DL architectures.

The optimisation results show a trade-off rather than a simple relationship in which more compression always produces better predictive performance. The 60% pruning target produced the strongest validation F1, while the 80% target produced the highest validation ROC-AUC. Quantisation produced the clearest directly measured efficiency change: storage fell by 34.10% and paired validation prediction time fell by 45.80%, while validation F1 changed by only 0.0045. These results are directionally consistent with previous work on compact and quantised ECG models (De Melo Ribeiro et al., 2022; Liu et al., 2024; Kim et al., 2022).

The participant-level test set contains only five independent participants, even though it contains 65,581 windows. Because consecutive windows overlap by 50% and windows from one participant are correlated, the number of windows should not be interpreted as the number of independent test observations. The small number of held-out participants therefore limits the strength of population-level generalisation claims.

A second limitation concerns the Wake/Sleep label. The target was created by matching each ECG window to the nearest available annotation within a ± 15 -minute tolerance. The annotation may not describe the participant’s state throughout the entire 10-second ECG window. The results should therefore be interpreted as classification under the implemented temporal-matching procedure rather than as clinical sleep-state measurement.

A third limitation is the preprocessing asymmetry between the model families. Traditional ML used nine features derived from 0.5–40 Hz filtered ECG, followed by training-only standardisation. The CNN and LSTM used waveform windows with per-window standardisation. The comparison is therefore a comparison of complete modelling pipelines rather than a controlled experiment in which every model receives exactly the same representation. This distinction prevents the observed CNN advantage from being overstated as a pure architecture

effect. The baseline models were also not subjected to a separate hyperparameter search in the reported experiment, so the very low Logistic Regression and SVM F1-scores should not be attributed to representation loss alone.

Class weighting was applied to the neural models. The paper does not report an equivalent class-weight adjustment for the Logistic Regression and Linear SVM baselines, which is another reason to interpret the baseline comparison as a pipeline comparison rather than a controlled test of model families.

Finally, efficiency was measured in the experimental software environment. The 34.10% storage reduction and $1.845\times$ measured speed-up provide evidence of software-level efficiency improvement, but they do not establish performance on a particular wearable processor. Actual deployment would require hardware-specific measurements of memory, inference time, acquisition and filtering cost, and real-time scheduling.

VI. CONCLUSION

This paper evaluated a compact ECG classification pipeline using the Wake/Sleep supervision available in the Dryad recordings. The experiment compared feature-based Logistic Regression and Linear SVM with waveform-based CNN and LSTM models, then selected the strongest baseline using validation data before applying pruning and post-training dynamic-range quantisation.

The lightweight CNN was the strongest baseline, achieving a validation F1-score of 0.7932 with 3,809 trainable parameters. Quantisation reduced measured model storage from 14.8789 KB to 9.8047 KB and reduced paired validation prediction time by 45.80%, while keeping validation F1 and ROC-AUC close to the Float32 model. The final quantised CNN achieved an F1-score of 0.7380 and ROC-AUC of 0.7736 on five held-out participants.

The main conclusion is deliberately narrow: within the implemented software experiment, a compact waveform CNN can distinguish the available Wake/Sleep labels while maintaining useful predictive performance after quantisation. The results do not establish broad population-level generalisation or real-world wearable deployment. Future work should use larger independent participant cohorts, more precise physiological-state annotations, sensitivity analysis of the temporal matching tolerance, more controlled representation comparisons, and hardware-specific evaluation of compressed models.

REFERENCES

- Ansari, Y., Mourad, O., Qaraqe, K., and Serpedin, E. (2023). “Deep learning for ECG Arrhythmia detection and classification: an overview of progress for period 2017–2023”. In: *Frontiers in Physiology* 14, p. 1246746. DOI: 10.3389/fphys.2023.1246746.
- Avula, V., Wu, K. C., and Carrick, R. T. (2023). “Clinical Applications, Methodology, and Scientific Reporting of Electrocardiogram Deep-Learning Models: A Systematic Review”. In: *JACC: Advances* 2.10, p. 100686. DOI: 10.1016/j.jacadv.2023.100686.

- Chen, S. W., Wang, S. L., Qi, X. Z., Samuri, S. M., and Yang, C. (2022). "Review of ECG detection and classification based on deep learning: Coherent taxonomy, motivation, open challenges and recommendations". In: *Biomedical Signal Processing and Control* 74, p. 103493. DOI: 10.1016/j.bspc.2022.103493.
- Choi, I. and Sung, W. (2023). "Single-Channel ECG-Based Sleep Stage Classification With End-to-End Trainable Deep Neural Networks". In: *2023 IEEE Engineering in Medicine and Biology Society Conference (EMBC)*, pp. 1–4. DOI: 10.1109/EMBC40787.2023.10340302.
- De Melo Ribeiro, H., Arnold, A., Howard, J. P., Shun-Shin, M. J., Zhang, Y., Francis, D. P., Lim, P. B., Whinnett, Z., and Zolgharni, M. (2022). "ECG-based real-time arrhythmia monitoring using quantized deep neural networks: A feasibility study". In: *Computers in Biology and Medicine* 143, p. 105249. DOI: 10.1016/j.combiomed.2022.105249.
- Haleem, M. S. and Pecchia, L. (2022). "A deep learning based ECG segmentation tool for detection of ECG beat parameters". In: *2022 IEEE Symposium on Computers and Communications (ISCC)*, pp. 1–4. DOI: 10.1109/ISCC55528.2022.9912906.
- Jia, Y., Pei, H., Liang, J., Zhou, Y., Yang, Y., Cui, Y., and Xiang, M. (2024). "Preprocessing and Denoising Techniques for Electrocardiography and Magnetocardiography: A Review". In: *Bioengineering* 11.11, p. 1109. DOI: 10.3390/bioengineering11111109.
- Kim, S., Chon, S., Kim, J.-K., Kim, J., Gil, Y., and Jung, S. (2022). "Lightweight Convolutional Neural Network for Real-Time Arrhythmia Classification on Low-Power Wearable Electrocardiograph". In: *2022 44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1915–1918. DOI: 10.1109/EMBC48229.2022.9871156.
- Liu, Y., Liu, J., Tian, Y., Jin, Y., Li, Z., Zhao, L., and Liu, C. (2024). "Pruned lightweight neural networks for arrhythmia classification with clinical 12-Lead ECGs". In: *Applied Soft Computing* 154, p. 111340. DOI: 10.1016/j.asoc.2024.111340.
- Petmezas, G., Stefanopoulos, L., Kilintzis, V., Tzavelis, A., Rogers, J. A., Katsaggelos, A. K., and Maglaveras, N. (2022). "State-of-the-Art Deep Learning Methods on Electrocardiogram Data: Systematic Review". In: *JMIR Medical Informatics* 10.8, e38454. DOI: 10.2196/38454.
- Prakash, A. J. and Atef, M. (2025). "A lightweight deep learning approach for patient-specific electrocardiogram beat classification using local and long-term dependencies". In: *Engineering Applications of Artificial Intelligence* 152, p. 110754. DOI: 10.1016/j.engappai.2025.110754.
- Safdar, M. F., Nowak, R. M., and Pałka, P. (2024). "Pre-Processing techniques and artificial intelligence algorithms for electrocardiogram (ECG) signals analysis: A comprehensive review". In: *Computers in Biology and Medicine* 170, p. 107908. DOI: 10.1016/j.combiomed.2023.107908.
- Stokes, K., Haleem, M. S., Pecchia, L., and Cappuccio, F. (2025). "24-hour physiological monitoring: Electrocardiogram, interstitial glucose, and ambulatory blood pressure". In: *Dryad Digital Repository*. DOI: 10.5061/dryad.qjq2bvqpk. URL: <https://doi.org/10.5061/dryad.qjq2bvqpk>.
- Stracina, T., Ronzhina, M., Redina, R., and Novakova, M. (2022). "Golden Standard or Obsolete Method? Review of ECG Applications in Clinical and Experimental Context". In: *Frontiers in Physiology* 13, p. 867033. DOI: 10.3389/fphys.2022.867033.
- Sun, H., Ganglberger, W., Panneerselvam, E., Leone, M. J., Quadri, S. A., Goparaju, B., Tesh, R. A., Akeju, O., Thomas, R. J., and Westover, M. B. (2020). "Sleep staging from electrocardiography and respiration with deep learning". In: *Sleep* 43.7, zsz306. DOI: 10.1093/sleep/zsz306.
- Xiao, Q., Lee, K., Mokhtar, S. A., Ismail, I., Pauzi, A. L. b. M., Zhang, Q., and Lim, P. Y. (2023). "Deep Learning-Based ECG Arrhythmia Classification: A Systematic Review". In: *Applied Sciences* 13.8, p. 4964. DOI: 10.3390/app13084964.

ECS750P/ECS753P/ECS754P– MSc Project Module
Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the final dissertation paper.

Student Name: Ayush Deshmukh

Project Title: Lightweight ECG Signal Classification Using Machine Learning and Deep Learning Models

Supervisor Name: Dr Tayyab Ahmad Ansari

1. Have you used Generative AI tools for your project?

YES

NO

If yes, please complete sections 2 and 3 and sign the declaration below.

If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
e.g. ChatGPT	5.6	OpenAI	https://chat.openai.com/
Claude	Version Available during project	Anthropic	https://claude.ai/new

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Example Table – please add your own

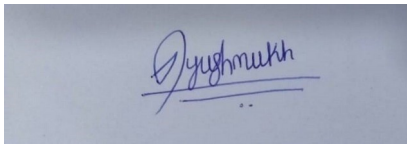
Project Section	Description	Justification
Initial project planning	Used ChatGPT, Claude to explore the ECG classification research area, clarify terminology, compare possible approaches, and help structure the initial project plan.	Used as a brainstorming and learning aid to understand the research landscape and identify areas requiring further investigation. Decisions about the final project direction were made by the student and discussed with the supervisor
Software development / coding support	Used GenAI tools to explain programming concepts, troubleshoot errors, suggest possible approaches to implementation and help understand code-related problems.	Used as a programming learning and debugging aid. The student reviewed, understood, tested and adapted the code and remained responsible for the implementation and results
Writing and editing	Used GenAI occasionally to identify grammatical issues, improve clarity, and suggest alternative wording or structure for parts of the project writing	Used for language and structural assistance rather than replacing the student's research or scholarly argument. The final content was reviewed and edited by the student.

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.

- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Jyushmukh' written in a cursive style, with a horizontal line underneath it.

Date: 19th august 2026